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Nonlinear viscoelastic, thermodynamically consistent, models for biological soft tissue

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Abstract The mechanical behavior of most biological soft tissue is nonlinear viscoelastic rather than elastic. Many of the models previously proposed for soft tissue involve ad hoc systems of springs and dashpots or require measurement of time-dependent constitutive coefficient functions. The model proposed here is a system of evolution differential equations, which are determined by the long-term behavior of the material as represented by an energy function of the type used for elasticity. The necessary empirical data is time independent and therefore easier to obtain. These evolution equations, which represent non-equilibrium, transient responses such as creep, stress relaxation, or variable loading, are derived from a maximum energy dissipation principle, which supplements the second law of thermodynamics. The evolution model can represent both creep and stress relaxation, depending on the choice of control variables, because of the assumption that a unique long-term manifold exists for both processes. It succeeds, with one set of material constants, in reproducing the loading–unloading hysteresis for soft tissue. The models are thermodynamically consistent so that, given data, they may be extended to the temperature-dependent behavior of biological tissue, such as the change in temperature during uniaxial loading. The Holzapfel et al. three-dimensional two-layer elastic model for healthy artery tissue is shown to generate evolution equations by this construction for biaxial loading of a flat specimen. A simplified version of the Shah–Humphrey model for the elastodynamical behavior of a saccular aneurysm is extended to viscoelastic behavior.

1 Introduction

The mechanical behavior of biological soft tissue is time dependent and nonlinear. A full description of the mechanical response of soft biological tissue to loads requires a nonlinear viscoelastic model, which is built on a firm theoretical and physical foundation, consistent with non-equilibrium thermodynamics. Existing models require a large number of material constants. A model that predicts the time-dependent behavior of soft tissue from a simple experiment is needed; preferably, the necessary empirical terms in the model should be few in number and should not be time dependent so that they are easy to obtain. Here, the time-dependent behavior is deduced from a time-independent model for the long-term behavior and a thermodynamic construction.

According to the small number of experiments available in the literature, the uniaxial stress–strain response of soft tissue exhibits hysteresis in a loading–unloading test. The hysteresis is believed to be only weakly dependent on the loading rate and therefore is only weakly dependent on frequency in the cyclic load testing. This belief seems to be founded on a set of experimental results performed by Fung on rabbit papillary muscle. Fung repeated in several papers that the loading and unloading uniaxial curves are almost strain-rate independent; however, his data at three different rates (as given in Fig. 2, 1972; Fig. 1a, 1973) only approximately supports this statement. A moderate sensitivity for tests under strain control was observed for articular cartilage by Woo et al. (1980) because in a loading and unloading test “the peak stresses increased slightly with increased strain rates”. The response may not be the same if load rather than strain is controlled. A study by Bennett et al. (1986) showed that the energy dissipated per cycle in a dolphin tendon was nearly independent of frequency; however, a close examination of their data for cyclic tests at 2.2 Hz but at different load amplitudes shows that the strain response is not independent of load rate. Even so, to simplify the

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modeling problem, some have assumed that the stress–strain behavior is time independent, or at least strain rate insensitive, and have proposed nonlinear elastic models for soft tissue.

Hyperelastic (i.e. time independent) models have commonly been used to represent the stress–strain relation of biological soft tissue in the special case of monotonically increasing loads (e.g. Skalak et al. 1975). Fung and his coworkers generated hyperelastic models for skin and for arterial tissue (Fung 1993, pp. 302–306). One is of the form of an exponential plus a constant and the historically later one is simply an exponential. The constants in this model depend on whether the material is being loaded or unloaded. Humphrey et al. (1992) suggested that a two-dimensional model of the form of the Fung exponential model can approximately represent the stress–stretch relation for canine epicardial tissue under monotonic loading, and Kyriacou and Humphrey (1996) fit the model to the stress–stretch data of Scott et al. (1972) for intracranial aneurysmal tissue. Humphrey and coworkers also developed such models for pleura and pericardium (e.g. Humphrey et al. 1987). Holzapfel et al. (2000), after a clear and detailed description of the physical structure of an artery wall, presented a two-layer hyperelastic model for the composite arterial tissue, which is orthotropic in each layer. The model accounts for residual stresses and requires three material parameters for each layer.

However, hyperelastic models cannot represent time-dependent processes such as creep or stress relaxation. Viscoelastic models proposed for soft tissue have included isothermal linear viscoelastic models in the form of spring and dashpot configurations, such as the modified standard linear solid used by Apter (1964) for the viscoelastic properties of strain-controlled loadings, either stress relaxation or harmonic loading, in aortic and pulmonary artery tissue of dogs. Unfortunately, the linear long-term relation between stress and strain predicted by most spring and dashpot models is not borne out by experiment.

Integral models, among the earliest proposed for viscoelastic behavior, require experimentally fitting a large number of empirical time-dependent functions. The difficulty in obtaining from experiment the unknown functions in these models has restricted their use to fit the mechanical behavior of tissue. A similar difficulty arises with the Coleman (1964) theory of simple materials with fading memory.

Soft tissue viscoelastic models using an infinite number of spring and dashpots have been of two major types, those which produce a continuous relaxation spectrum and those intended to represent the tissue as a network of fibers. A nonlinear Kelvin–Voigt type model was constructed by Viidik (1968) using a sequence of springs of different lengths so that more springs are stretched as the material deforms, to account for the increasing stiffness of artery tissue with stretch. Lanir (1983) built a model from the properties of the structural components of the tissue, the elastin and collagen.

A probability distribution function is required to describe the density of each type of fiber. The behavior of a fiber is determined by a fading memory function, which is more general than a spring and dashpot model, rather than a hyperelastic function. This idea is similar to network models proposed for paper sheets (e.g. Perkins 1983) except that the paper fiber network models in addition require a bond at the positions where the fibers cross. The complexity of the Lanir type models and the difficulty of determining the fading memory functions limit their usefulness. Further, the constituents may behave differently in isolation than they do in the composite tissue. Other investigators in the middle of the 20th century examined the dynamic moduli under harmonic loads of different frequencies. Such analyses assume linear viscoelastic behavior.

Some nonlinear empirical viscoelastic models have also been proposed such as that analogous to a rubber model by Fung (1972). While this Fung model successfully fits the stress relaxation data, it does not fit creep data. Decraemer et al. (1980) further developed Fung's idea and pointed out its limitation to a small range of frequencies. They suggested that the viscoelastic response was due to friction between the structural fibers of the tissue and their surroundings. They assumed a large number of fibers of different initial lengths that were embedded in a gelatin-like matrix. As the deformation of the specimen increases, more of these fibers are stretched to represent the experimentally observed nonlinear stress–strain response.

A constitutive model devised by Tanaka and Yamada (1990) specifically to represent the stress–strain hysteresis in the loading–unloading of blood vessels avoids the elaborate spring and dashpot constructions of others and is based on evolution equations for the internal variables such as those used in models for viscoplasticity. But similar to those models, their model assumes that the total strain splits into the sum of an elastic and a viscoelastic part. Most of the material coefficients were found by numerically fitting the hysteresis curve, and may depend on the chosen strain range, so that the model as presented is not predictive.

A more recent strategy to build a viscoelastic model has been to write the Helmholtz energy as the sum of a long-term hyperelastic energy and a viscous energy (e.g. Holzapfel 2000; Bonet 2001). The stress is then a sum of an elastic component and a dissipative component that tends to zero over time. The Bonet (2001) large-strain viscoelastic model based on splitting the deformation gradient into elastic and viscous components and on a generalized Maxwell model for the viscous component was extended to artery tissue by Holzapfel et al. (2002). This three-dimensional viscoelastic model for artery tissue, viewed as a two-layer (media and adventitia) cylinder with collagen reinforcing fibers, which explicitly shows the long-term behavior, uses the Spencer hyperelastic composite construction presented in Holzapfel (2000). The isochoric elastic function is taken as the sum of a neo-Hookean model and an exponential model for

the collagen from Holzapfel et al. (2000). Their goal was to achieve a weakly frequency-independent hysteresis response, which they take to mean that the area between the loading and unloading uniaxial curves is nearly independent of frequency. Their sample dissipative energy uses five Maxwell models with different relaxation times for each to cover a four-decade time domain of 0.01–100 Hz. Another five factors are computed to make the dissipation energy nearly constant over the design frequency range. Therefore, this model requires a large number of empirical dissipation constants in addition to those for the elastic terms.

Here, a nonlinear viscoelastic model is built from non-equilibrium thermodynamic foundations that allow the evolution equations for dissipation behavior to be derived from an experimental long-term model, rather than be assumed. As in nonlinear dynamics, the long-term states organize the time-dependent transient behavior. The construction succeeds in modeling solid behavior, where other non-equilibrium thermodynamics theories fail, because thermoelastic, thermoviscoelastic and thermoviscoplastic behaviors are unified within this mathematical structure (Haslach 1997; Haslach and Zeng 1999; Haslach 2002). This construction when applied to the plastic deformation of metals (Haslach 2002) produces a viscoplastic model in which a threshold stress plays no role.

Data describing the long-term behavior of the biological material is much easier to collect than determining the time-dependent empirical functions required in classical integral models of viscoelasticity. The long-term behavior is not time-dependent and so is represented, in general, by a nonlinear hyperelastic model. The tissue behavior is assumed homogeneous, in the sense that there are no spatial gradients in the material.

The general construction of the evolution equations is first described. This construction is then carried out for several possible models of the long-term behavior of soft tissue. The resulting predictions for uniaxial creep, stress relaxation, monotonic and cyclic loading are compared to experimental data in the literature, when it exists. The application based on the Holzapfel et al. (2000) model to creep of a flat membrane under biaxial loading is outlined, as is a method to describe uniaxial stretch and isentropic temperature evolution due to dissipation. The Shah–Humphrey (1999) elastodynamical model for a saccular aneurysm is extended to viscoelastic behavior.

2 General nonlinear thermoviscoelastic model

Rate-dependent responses, including creep, stress-relaxation, monotonic and cyclic loading, are non-equilibrium processes. Cyclic loading is particularly important in the study of tissue such as that in artery walls or aneurysms loaded by the periodic blood pressure and in lung tissue subjected to periodic air flow. The model must be consistent with non-equilibrium thermodynamics. The structure of non-equilibrium thermodynamics is

not generally agreed upon, and existing theories are controversial. Classical non-equilibrium thermodynamics is based on the Onsager relations, the local reversibility of time, and the Curie principle forbidding coupling of variables with different tensorial types. All of these postulates have been challenged by many authors (e.g. Truesdell 1984). Rational thermodynamics uses the balance laws of continuum mechanics and the second law in the form of the Clausius–Duhem inequality to derive constraints on admissible constitutive relations, but does not explicitly construct a mathematical structure in which non-equilibrium thermodynamics can be placed and studied. The generalized thermodynamics of Prigogine and others requires an assumption of local equilibrium so that entropy and other concepts can be defined as for equilibrium states. It is primarily based on statistical mechanics and so has been most successfully applied to gases but not to solids. Its criteria for stability for a non-equilibrium process have been criticized as ad hoc by several researchers. The model, which depends on the state being close to equilibrium, cannot deal well with nonlinear non-equilibrium processes.

The thermodynamically consistent, nonlinear thermoviscoelastic model is based on a generalized energy function which makes explicit the conjugate relationships between state and control variables and for which generalized, nonlinear Onsager type relations can be defined, nonlinear processes can be modeled, and non-equilibrium processes described (Haslach 1997). A maximum dissipation hypothesis produces a thermodynamically consistent system of evolution equations for the state variables, which is the constitutive model for the thermoviscoelastic behavior.

The construction of the model requires a set of pairs of conjugate thermodynamic variables and relations defining the long-term manifold. Some of the variables may be internal variables. The long-term behavior is assumed to be described by a time-independent energy function, $E(y_1, \dots, y_n)$ of the control variables. The variables, such as the stress, strain, temperature, and entropy, are separated into pairs of state variables, x_i , and the control variables, y_i , so that pairs are conjugates in the sense that $\partial E / \partial y_i = x_i$ at thermodynamic equilibrium. For example, temperature and entropy or stress and strain are conjugate pairs. In creep, stress is the control and strain is the state variable; their roles are reversed in classical stress relaxation. The conjugacy equations define the long-term states.

To ensure that all types of loading can be modeled, it is assumed that the long-term manifold is well defined for each pair of conjugate variables in the following sense. For example, consider the conjugate pair stress and strain. Assume that a creep test under a control stress of σ_c asymptotically approaches a strain of ϵ_c . The long-term manifold is well defined and unique for this conjugate pair of variables if the stress in a stress relaxation test under a constant control strain of ϵ_c asymptotically approaches σ_c .

A generalized energy function, E^* , is defined on both equilibrium and non-equilibrium states and preserves the conjugate relations between control and state variables. Also, the long-term manifold, M_e , in $\mathbb{R}^{2n}$ defined by the energy function, $E(y_1, \dots, y_n)$ of the control variables such that $\partial E / \partial y_i = x_i$, is recovered through a zero gradient condition on E^* .

Definition 2.1. A generalized energy function defining a thermodynamic system is a smooth function $E^* : \mathbb{R}^{2n} \rightarrow \mathbb{R}$ such that

$$\frac{\partial E^*}{\partial y_i} = x_i \quad \text{for } i = 1, \dots, n. \quad (1)$$

The long-term submanifold, M_e , is described locally by

$$\frac{\partial E^*}{\partial x_i} = 0 \quad \text{for } i = 1, \dots, n. \quad (2)$$

Also, the restriction of E^* to M_e is the energy function, E , of the control variables up to a sign.

The simplest form of a generalized energy function is, for $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \dots, y_n)$,

$$E^*(\mathbf{x} : \mathbf{y}) = \psi(\mathbf{x}) + \sum_{i=1}^n x_i y_i, \quad (3)$$

where $\psi(\mathbf{x})$ is obtained from the Legendre transform of $E(y_1, \dots, y_n)$. The energy function, ψ , for a biological material could be one of the several hyperelastic models proposed for the stress-strain behavior at large deformations.

Non-equilibrium relaxation processes are defined in terms of new variables which are nonlinear generalizations of the affinities and fluxes used in classical linear non-equilibrium thermodynamics. Nonlinear affinities, X_i , defined as derivatives of the generalized energy function with respect to state variables, $X_i \equiv \partial E^* / \partial x_i = \partial \psi / \partial x_i + y_i$, for $i = 1, \dots, n$, act as generalized forces driving the system to its long-term state. Dissipation is measured as the rate of change of the generalized energy function. Let the function E'^* be the function E^* written in terms of the affinities. The expression given by the chain rule

$$\frac{dE^*}{dt} = \sum_{i=1}^n \frac{\partial E'^*}{\partial X_i} \dot{X}_i \quad (4)$$

is the dot product of the gradient of E'^* with respect to the affinities and the vector $\vec{X}$. It can be viewed as the directional derivative in the direction $\vec{X}$; the directional derivative is maximal in the direction of the gradient.

The fundamental non-equilibrium process postulated in the viscoelastic class considered here is called a gradient non-equilibrium process. For fixed control variables, $\mathbf{y}$, perturbed from equilibrium, the relaxation path, $\gamma(t)$, as a function of time in control-affinity space is $\gamma: \mathbb{R} \rightarrow \mathbb{R}^{2n}$ having coordinates $(\mathbf{y}, X_1(t), \dots, X_n(t))$ proceeds in the direction of maximum dissipation at a speed defined by a coefficient of proportionality, called

the relaxation modulus, which depends on the material. The tangent to the relaxation path is proportional to the negative of the gradient of $E'^*(\mathbf{y}, \mathbf{X})$.

A non-equilibrium process is driven by the approach of these nonlinear affinities to zero at the long-term state, as in linear non-equilibrium thermodynamics. The physical hypothesis assumed to supplement the second law of thermodynamics is that non-equilibrium relaxation processes proceed in state space in the direction of maximum dissipation of the generalized energy function with respect to the affinities, so that the non-equilibrium path is well defined. The speed at which the response follows this path is determined by the relaxation modulus.

Definition 2.2. A gradient relaxation process is one for which the evolution of the affinities, X_i , under fixed control variables is given by

$$\frac{dX_i}{dt} = -k \frac{\partial E'^*}{\partial X_i}, \quad \text{for } i = 1, \dots, n, \quad (5)$$

for a positive material function, k , called the *relaxation modulus*.

Each time derivative defines a component of the velocity vector, the tangent vector to the process path, in the state space for E'^* . The evolution rate depends on the distance from the manifold of long-term states as measured by the X_i . A relaxation process evolves towards the long-term manifold, where each $X_i = 0$.

Relaxation is used in the broad sense of any process occurring while the control variables are held fixed. The control variables of a system originally in equilibrium are suddenly perturbed, so that the substance has the old state variables, but new control variables. Viscoelastic relaxation is a non-equilibrium process whose long-term behavior, or equilibrium manifold, is modeled by linear or nonlinear elasticity.

A forced process is a non-equilibrium process in which the control variables vary with time. The rather vague second law of thermodynamics does not determine the exact evolution of a non-equilibrium process; it must be augmented by at least one additional assumption. The gradient relaxation process in the case that the control variables are fixed over time is generalized for arbitrary non-equilibrium processes so that any admissible process is locally a maximum dissipation process.

Definition 2.3. Each non-equilibrium process is locally a maximum dissipation process, in the sense that it is a gradient relaxation process over each very short-time interval.

This condition can be viewed as the constitutive restriction defining a class of thermoviscoelastic materials.

By a calculation given in Haslach and Zeng (1999) for long-term manifold represented by a function with non-singular Jacobian (most solids), a thermoviscoelastic maximum dissipation process is represented by a system of first-order nonlinear ordinary differential equations in terms of the state variables, x_i , and the control variables, y_i ,

$$\begin{aligned}
\begin{bmatrix} \frac{dx_1}{dt} \\ \frac{dx_2}{dt} \\ \vdots \\ \frac{dx_n}{dt} \end{bmatrix} &= -k \begin{bmatrix} \frac{\partial^2 \psi}{\partial x_1^2} & \frac{\partial^2 \psi}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 \psi}{\partial x_1 \partial x_n} \\ \frac{\partial^2 \psi}{\partial x_2 \partial x_1} & \frac{\partial^2 \psi}{\partial x_2^2} & \cdots & \frac{\partial^2 \psi}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 \psi}{\partial x_n \partial x_1} & \frac{\partial^2 \psi}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 \psi}{\partial x_n^2} \end{bmatrix}^{-2} \\
&\quad \begin{bmatrix} -y_1(t) + \frac{\partial \psi}{\partial x_1} \\ -y_2(t) + \frac{\partial \psi}{\partial x_2} \\ \vdots \\ -y_n(t) + \frac{\partial \psi}{\partial x_n} \end{bmatrix}. \quad (6)
\end{aligned}$$

Again, the relaxation constants, k , depend on the particular choice of the control variables. All material parameters except the relaxation modulus appear in the long-term behavior model so that only a small number of material coefficients must be found experimentally. None of these coefficients are time-dependent. The construction can be applied to any thermostatic constitutive model that represents the long-term behavior of a viscoelastic material, for example, when the long-term manifold is represented by any of the classical hyperelastic models for large deformations, such as the neo-Hookean, the Mooney-Rivlin, and the Ogden (1972) models for rubber or by any of the hyperelastic models proposed for soft tissue. This construction also generates thermoviscoplastic metal models by letting the relaxation modulus be a function of the time-dependent internal state variables to account for the changing dissipation due to the variations in dislocation structure created by the load (Haslach 2002).

To produce large deformation, three-dimensional evolution models using the construction presented here, a choice must be made of a strain measure and a stress that are conjugate in the sense underlying the evolution equations. One possible pair is the Green strain, $\mathbf{E} = 0.5(\mathbf{C} - \mathbf{I})$, and the second Piola-Kirchhoff stress, $\mathbf{S}$, since $\mathbf{S} = \partial \psi / \partial \mathbf{E}$. The tensor $\mathbf{C} = \mathbf{F}^T \mathbf{F}$, where $\mathbf{F}$ is the deformation gradient tensor. Recall that by objectivity and the polar decomposition $\psi(\mathbf{F}) = \psi(\mathbf{C}) = \psi(\mathbf{E})$. Another possibility is the deformation tensor, $\mathbf{F}$, and the first Piola-Kirchhoff stress, $\mathbf{P}$.

2.1 Linear uniaxial long-term manifolds

Several classical spring and dashpot, uniaxial, linear viscoelastic models are recovered from this construction when the long-term manifold is linear elastic. The choice of the relaxation modulus, k , determines the particular model. The general forms of creep and stress relaxation are first derived from a linear long-term manifold.

The case that the state variable is the stress, σ , and the control variable is the strain, ε , has energy function $\psi(\sigma) = 0.5 \sigma^2 / E$ for a linear elastic long-term manifold. The evolution equation 6 for the state variable is then

$$\dot{\sigma} = -k_\sigma E^2 \left[-\varepsilon(t) + \frac{\sigma}{E} \right]. \quad (7)$$

In the dual case when the state variable is the strain, ε , and the control variable is the stress, σ , the linear elastic energy function for the long-term manifold is $\psi(\varepsilon) = 0.5 E \varepsilon^2$. The evolution of ε is predicted by Eq. 6 to be

$$\dot{\varepsilon} = -k_\varepsilon E^{-2} [-\sigma(t) + E \varepsilon]. \quad (8)$$

The Kelvin-Voigt model of a spring and dashpot in parallel is obtained when $k_\varepsilon = E^2/c$, where c is the dashpot viscous constant and E is the spring constant. The Kelvin-Voigt is locally maximum dissipative when strain is the state variable because Eq. 8 agrees with the Kelvin-Voigt differential equation, $c \dot{\varepsilon} + E \varepsilon = \sigma$, where the dot indicates the time derivative. Apter and Marquez (1968) state that collagen by itself appears to obey a Kelvin-Voigt model and does not seem to stress relax. The Kelvin-Voigt model does not admit stress relaxation because there is no $\dot{\sigma}$ - term. If the strain is held fixed so that $\dot{\varepsilon} = 0$, the stress remains unchanged, $\sigma = E \varepsilon$. Therefore, to represent this fact, take $k_\sigma = 0$ in Eq. 7.

The *standard linear solid* is a spring with constant, E_1 , in parallel with a Maxwell model, which is itself a dashpot in series with a second spring. The spring constant and dashpot viscosity in the Maxwell model are E_2 and c , respectively. The differential equation that the applied stress, σ , and the total strain, ε , must satisfy is

$$E_2 \sigma + c \dot{\sigma} = E_1 E_2 \varepsilon + c(E_1 + E_2) \dot{\varepsilon}. \quad (9)$$

In stress relaxation for fixed ε , after putting $\dot{\varepsilon} = 0$ in Eq. 9 and integrating, the stress is

$$\sigma(t) = \sigma(0) \exp(-kt) + \varepsilon E_1 (1 - \exp(-E_2 t/c)). \quad (10)$$

Over the long term, the stress-strain relation becomes linear, $\sigma = \varepsilon E_1$. Empirical constants for this model applied to artery tissue were obtained by Apter et al. (1966) from stress relaxation tests that reached a steady state within 2 min. Their goal was to relate the constants to the proportion of the constituents (e.g. collagen, elastin, muscle) of the aorta so the constituents could be determined from in vivo pressure curves. The method did not achieve conclusive results.

Creep is predicted by Eq. 9 with $\dot{\sigma} = 0$. The solution is

$$\varepsilon(t) = \frac{\sigma}{E_1} + \exp\left(\frac{-E_1 E_2 t}{c(E_1 + E_2)}\right) \left(\varepsilon(0) - \frac{\sigma}{E_1} \right). \quad (11)$$

Creep again has a linear elastic long-term manifold with $\sigma = E_1 \varepsilon$. The long-term manifold predicted by Eq. 9 is therefore the same in both creep and stress relaxation.

The evolution form of stress relaxation of the standard linear solid is given by Eq. 6 with the state variable, σ , and fixed ε with long-term manifold defined by $\psi(\sigma) = 0.5 \sigma^2 / E_1$. The result is given by Eq. 7 with the relaxation modulus $k_\sigma = E_2 / (E_1 c)$.

$$\dot{\sigma} = -\frac{E_1 E_2}{c} \left(-\varepsilon + \frac{\sigma}{E_1} \right). \quad (12)$$

This equation agrees with setting $\dot{\sigma} = 0$ in Eq. 9. Therefore, the stress relaxation results reported by Apter et al. (1966), who described them by the standard linear solid (Eq. 9), can also be represented by the viscoelastic construction proposed here.

Creep of the standard linear solid is obtained as well from Eq. 8 when strain is the state variable by taking $k_e = E_1^2 E_2 / c(E_1 + E_2)$. The long-term behavior of the standard linear solid is $\sigma = E_1 \varepsilon$ so that $\psi(\varepsilon) = 0.5 E_1 \varepsilon^2$.

$$\dot{\varepsilon} = \frac{E_2}{c(E_1 + E_2)} (-\sigma + E_1 \varepsilon). \quad (13)$$

This agrees with the equation obtained by setting $\dot{\sigma} = 0$ in Eq. 9.

The standard linear solid is not, in general, locally maximum dissipative because Eq. 9 does not agree with the maximum dissipation evolution equation from Eq. 6 under stress control,

$$\dot{\varepsilon} = \frac{E_2}{c(E_1 + E_2)} (-\sigma(t) + E_1 \varepsilon), \quad (14)$$

when the control variable $\sigma(t)$ is not constant; the evolution Eq. 14 does not include the $\dot{\varepsilon}$ -term which appears in Eq. 9.

3 Uniaxial nonlinear viscoelastic models for biological tissue

A different evolution equation is induced by each possible model for the long-term manifold and relaxation modulus. To verify that the construction of a nonlinear viscoelastic model given in the previous section can in fact apply to biological tissue, the technique is applied to two different Fung and the Skalak et al. hyperelastic models. None of the models taken here as examples for the long-term behavior was originally proposed for that purpose. In each case, the long-term model is defined, then the evolution equations are developed, and finally, using available data for the constants in the long-term model, the prediction of the evolution equations is compared, if possible, to the published experimental results.

3.1 Long-term manifold represented by the Tanaka–Fung initial elastic model

The most complete published experimental information on the nature of the long-term manifold for soft tissue comes from Fung (1972) in which it is shown that the stress–stretch relation in the long term has the same qualitative shape as commonly given experimental stress–stretch relations for soft tissue, such as Scott et al. (1972).

Fung (1972) developed an expression for tissue stress, $T(\lambda)$, as a function of stretch, λ , the current length divided by the reference length, using the fact that in a

certain range the experimental curve for $dT/d\lambda$ could be fit by the straight line $dT/d\lambda = \alpha(T + \beta)$. The experimental curve is obtained under stretch control by applying instantaneous steps, or jumps, in stretch to the specimen. The stress, which is the corresponding conjugate state variable, is the force divided by the reference area. Fung obtained

$$T(\lambda) = (T^* + \beta) \exp(\alpha(\lambda - \lambda^*)) - \beta, \quad (15)$$

where (λ^*, T^*) is a point on the stress–strain curve, and α and β are the material constants. Here Fung's notation is used so that the reader can more easily compare the results obtained from the proposed model to Fung's various publications.

The initial elastic stress, $T^{(e)}(\lambda)$, the immediate instantaneous elastic response to a step function of stretch, is approximately equal to the tensile stress induced under a high rate of loading (Fung 1972), so that $T^{(e)}(\lambda)$ may be modeled by Eq. 15.

Following a traditional strategy of using rubber models as guidance for modeling soft tissue, Fung assumed the time-dependent stress response factors to be a product of a function, which depends only on time and one function depending on stretch alone. Rubber stress relaxation was shown by Guth et al. (1946) to be fit closely by such a factored model, but factorization does not hold for creep of rubber. Stress relaxation of soft tissue was represented by Fung (1972) using $T = G(t)T^{(e)}(\lambda)$, where $G(t)$ is an empirical function.

Fung (1972) generalized this model to linear viscoelasticity by replacing the product of the two functions, G and $T^{(e)}$, with their convolution integral. The stress relaxation is then $T(t) = \int_{-\infty}^t G(t - \tau) \dot{T}^{(e)}(\lambda(\tau)) d\tau$. According to Fung, this viscoelastic model for the stress response to variations in stretch was said to be a quasi-linear model because a linear viscoelastic relation holds between T and $T^{(e)}$ but the function $T^{(e)}(\lambda)$ is nonlinear in λ .

Unfortunately, in general, the experimental determination of $G(t)$ is not easy. The approximation $G(t) = c \ln t + d$, classically used for the creep of metals under low stresses, has frequently been proposed for biological tissue but does not hold for arteries (Tanaka and Fung 1974). Small strain creep in heart tissue does seem to follow a logarithmic curve for up to 10 s, and $G(t)$ is independent of temperature, at least when the ambient temperature range is from 5 to 37°C (Fung 1993, pp. 404–405). Fung (1972) instead took

$$G(t) = \frac{1 + c[\varepsilon(t/\tau_2) - \varepsilon(t/\tau_1)]}{1 + c(\ln \tau_2 - \ln \tau_1)}, \quad (16)$$

where $\varepsilon(t/\tau_i) = \int_0^{t/\tau_i} (1/\tau) \exp(t/\tau) d\tau$, c is a constant, and the τ_i are the bounds on a range of time in which an ad hoc normalization factor applies.

Therefore, because $T(t, \lambda) = G(t)T^{(e)}(\lambda)$, the long-term manifold is assumed for the construction here to be $T(\lambda) = G(\infty)T^{(e)}(\lambda)$. Thus, according to Fung's experimental data, the long-term manifold is just a scalar

multiple of the initial elastic response. To qualitatively reproduce the Fung experimental data by Eq. 6, the Helmholtz energy, $\psi(\lambda)$, is assumed to satisfy

$$T(\lambda) = \frac{d\psi}{d\lambda} = G(\infty)T^{(e)}(\lambda) = \left[1 + c \ln\left(\frac{\tau_2}{\tau_1}\right)\right]^{-1} T^{(e)}(\lambda), \quad (17)$$

using Eq. 16 and the fact that $\lim_{t \rightarrow \infty} \varepsilon(t/\tau_i) = 0$. Note that $G(\infty) < 1$ if $\tau_2 > \tau_1$. The nonlinear viscoelastic evolution equation 6 for the stretch, λ , as the state variable and the stress, σ , as the control variable is

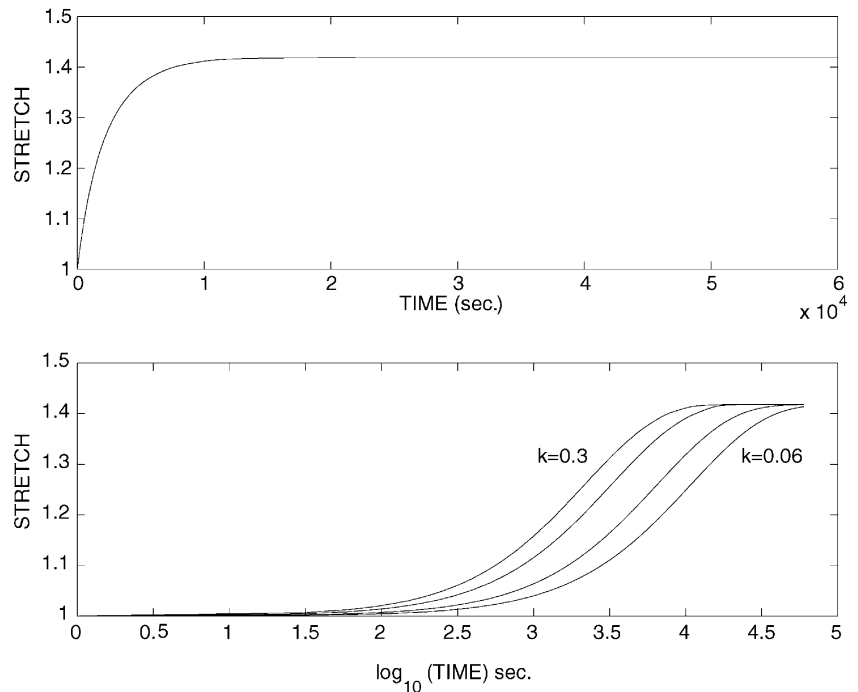
$$\dot{\lambda} = -k[\alpha G(\infty)(T^* + \beta) \exp(\alpha(\lambda - \lambda^*))]^{-2} \times \{-\sigma(t) + G(\infty)[(T^* + \beta) \exp(\alpha(\lambda - \lambda^*)) - \beta]\} \quad (18)$$

Tanaka and Fung applied the model (15) to their data (Table 1, 1974) for the creep of a circumferential segment of the arch region of a dog's carotid arterial tissue. The tissue produces different constants at different positions along the artery. Tanaka and Fung (1974) say that the quasi-linear relationship cannot be expected to hold for the full range of inelastic large deformation behavior. This expression is valid for stresses in the range of 20–60 kPa (3–9 psi) (Fung 1993, p. 328); Tanaka and Fung specify $T > 200$ g/cm². The constants in the model for the long-term manifold are chosen from their experimental data as $\lambda^* = 1.6$, $T^* = 535$ g/cm², $\alpha = 0.57$ and $E_o \equiv \alpha\beta = 910$ g/cm². From their Table 7, $c = 0.0424$, $\tau_1 = 0.367$ s, and $\tau_2 = 434$ s for the circumferential arch so that $G(\infty) = 0.7692$. For these values, Eq. 15 gives negative stresses if the stretch is $1 \leq \lambda < 1.1$ and so to test this model for creep, the creep stress must be chosen in the allowable stress range for which the long-term manifold is defined.

To generate numerical results for Eq. 18, the long-term manifold is therefore taken to be $G(\infty)T(\lambda) = 0.7692T(\lambda)$, where $T(\lambda)$ is given by Eq. 15. The numerical computations were carried out with an adaptive step-size low order Runge-Kutta Fehlberg method (e.g. MATLAB ode23). Figure 1a shows the creep versus time for $k = 0.3$ and Fig. 1b shows the creep curves for $k = 0.06, 0.1, 0.2, 0.3$, obtained when the abscissa is the log base 10 of time. The initial condition is $\lambda(0) = 1$ and the stress is $\sigma = 250$ g/cm². The behavior is controlled by the relaxation modulus, k . The larger the value of k , the further to the left the creep curve moves because the material relaxes faster to its long-term state. Even larger values of k cause the curve to oscillate about the long-term manifold. The curves in Fig. 1b have the same qualitative S-shape as the experimental data of Tanaka and Fung (1974, Fig. 12), although the magnitude is much greater because the creep stress was taken large enough, 250 g/cm², to be in the range of validity of the long-term manifold. Their Fig. 12 was generated under a much smaller creep stress of about 22 g/cm². The curve turns sharply up at 60,000 s similarly to the result of Tanaka and Fung. The important outcome is that this model captures the qualitative behavior under creep that the Fung quasi-linear model could not fit (Fung 1993, p. 287). Fung said that the quasi-linear theory does not apply well to longer term creep. This failure is consistent with the fact that the factorization model also does not fit the creep of rubber (Guth et al. 1946).

The corresponding stress relaxation evolution equation model Eq. 6 has stress as the state variable. To construct it, solve Eq. 15 for the stretch, λ , and assume there is an energy function φ (which is a Legendre transform of the Helmholtz energy $\psi(\lambda)$) such that

Fig. 1 Creep response from the Tanaka–Fung long-term model with constants $\lambda^* = 1.6$, $T^* = 535$ g/cm², $\alpha = 0.57$, $\beta = 910/\alpha$ g/cm², and $G(\infty) = 0.7692$, under a stress of $\sigma = 250$ g/cm² and $\lambda(0) = 1$. The solution is by MATLAB ode23. **a** Stretch versus time for $k = 0.3$. **b** Stretch versus $\log_{10}$ of time for $k = 0.06, 0.1, 0.2, 0.3$. The curve on the right corresponds to $k = 0.06$ and the curves to the left correspond to the increasing values of k as move to the left in order



$$\lambda \frac{d\phi}{dT} = \lambda^* + \frac{1}{\alpha} \left[\ln \left(\frac{[T/G(\infty)] + \beta}{T^* + \beta} \right) \right].$$

Then $d^2\phi/dT^2 = 1/[\alpha (T + \beta G(\infty))]$ and the evolution equation 6 is

$$\dot{T} = -k_T [\alpha(T + \beta G(\infty))]^2 \times \left\{ \lambda^* + \frac{1}{\alpha} \left[\ln \left(\frac{[T/G(\infty)] + \beta}{T^* + \beta} \right) \right] - \lambda \right\}, \quad (19)$$

for a fixed stretch λ . The initial condition is the initial stress.

The stress relaxation, of the stress plotted against $\log_{10}(t)$ (Fig. 2), shows an S-shaped curve, when $k_T = 0.00001, 0.00002, 0.00003$ or 0.00004 . The fixed stretch is 1.4, therefore the initial stress is $T^{(e)}(\lambda) = 305.35 \text{ g/cm}^2$. This S-shape is hinted at in the experimental relaxation curves $G(t)$ in Tanaka and Fung (1974, Fig. 8), which show a slower initial relaxation, a faster relaxation and finally a slower relaxation (see also Fung 1993, Fig. 7.6:4). The larger the value of k_T , the faster the relaxation so that the value of k increases as one moves from a curve to the one to its left in Fig. 2.

Because the Tanaka–Fung model is only defined if $T > 200 \text{ g/cm}^2$, the construction using it for the long-term manifold cannot be applied to produce a stress–strain curve which begins at zero stress.

3.2 Long-term manifold represented by the Fung exponential model

Fung later modified his hyperelastic constitutive model Eq. 15 by dropping the term added to the exponential; the resulting, purely exponential form seemed to better fit the data over a full range of stress. It has been applied

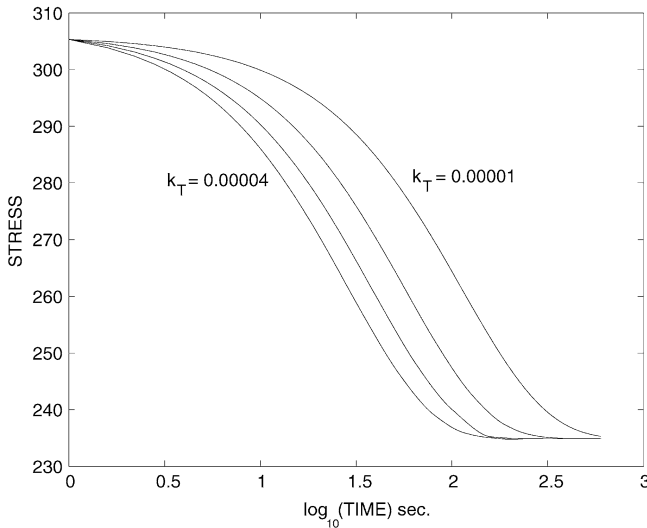


Fig. 2 Stress relaxation from the Tanaka–Fung long-term model with constants $\lambda^* = 1.6$, $T^* = 535 \text{ g/cm}^2$, $\alpha = 0.57$, $\beta = 910/\alpha \text{ g/cm}^2$, and $G(\infty) = 0.7692$ under a stretch of $\lambda = 1.4$, initial stress is $T^{(e)}(\lambda) = 305.35 \text{ g/cm}^2$, and $k = 0.00001, 0.00002, 0.00003$ or 0.00004 . The solution is by MATLAB ode23

to both arterial and lung membrane tissue by Fung and coworkers. The strain energy density for uniaxial loading is of the form, $\psi(E) = c[\exp(c_1 E^2) - 1]$, where the Green strain $E = (\lambda^2 - 1)/2$, for the stretch λ . For this strain energy, the second Piola–Kirchhoff stress, S , is given by

$$S = \frac{d\psi}{dE} = 2cc_1 E \exp(c_1 E^2). \quad (20)$$

Tong and Fung (1976) emphasize that the exponential model, applied to rabbit abdominal skin, requires different empirical constants in loading and unloading. This lack of universality is a consequence of their use of a hyperelastic rather than a viscoelastic model. Most applications of this model have been to fitting monotonic stress–strain tests. Here, the construction is instead applied to $\psi(E) = c \exp(c_1 E^2)$ taken as the model for the long-term behavior, rather than as the traditional stress–stretch relation, even though no data exists applying it to the long-term behavior. In stress control, at a given stress S , the instantaneous strain E is less than its value on the long-term manifold. In strain control at a given E , the instantaneous stress, S , is greater than the value on the long-term manifold. An estimate is needed to relate available stress–strain data to the long-term behavior.

Because the time-dependent response of this model is given by the differential equation 6 rather than in closed form, experimental data cannot be fit by finding the material constants by techniques such as least squares. Much error would be introduced in the alternative attempt to compute the derivatives to printed experimental graphs given in the literature without having the original raw data. For the purpose of illustrating the construction of the evolution Eq. 6, the single data point that Steiger et al. (1989, Fig. 4) provide for the long-term stress relaxation behavior of an aneurysm will be used to estimate the long-term manifold from the stress–strain data. Since no experimental data is given for the long-term behavior at all strains, it is assumed that the long-term stress–strain relation is proportional to the stress–strain relation given in strain control. The long-term stress under stress relaxation from 1,250 kPa at $E = 0.34$ is about $S = 430 \text{ kPa}$. Then, to represent the long-term manifold, the Fung stress–strain relation is reduced by $430/1250 = 0.344$. Therefore, the long-term stress–strain relationship is obtained from the Kyriacou and Humphrey (1996) fit of the Fung model to the Scott et al. (1972) aneurysm tissue data and the reduction factor of 0.344 from Steiger.

The evolution Eq. 6 for Green strain as a state variable and the second Piola–Kirchhoff stress, S , as a control variable is

$$\dot{E} = -k_E [c(2c_1 + 4c_1^2 E^2) \exp(c_1 E^2)]^{-2} \times [2cc_1 E \exp(c_1 E^2) - S(t)]. \quad (21)$$

The creep strain versus time graph has a shape which appears similar to a logarithmic curve. That it is not

logarithmic is verified by plotting the strain against $\log(t)$. Then an S-shaped curve is produced as in the Tanaka and Fung (1974) data. The S-shape of the curve is not affected by the magnitude of the applied stress. As k_E is increased, the curve approaches its long-term value faster so that the initial portion is steeper.

This model (21), when $S(t) = st$ for constant stress rate s , predicts the stress–stretch curve of Fig. 3 under stress control for $s = 0.05$. The construction here captures the loading–unloading hysteresis that, by its very nature, the hyperelastic model cannot. In this representation, the same value of k_E works for both loading and unloading in contrast to the Tanaka–Fung model, which required two sets of material constants to predict loading–unloading hysteresis. In stress control, this model predicts a short initial concave down portion unless k_E is taken larger than in Fig. 3, where $k_E = 0.00001$. Smaller k_E produce the S-shaped stress–stretch curve, which is typical of rubber. Therefore, the numerical experimentation suggests that perhaps the magnitude of the relaxation modulus is one factor that distinguishes rubber from soft tissue.

The inverse function for $S = d\psi/dE = 2cc_1E \exp(c_1E^2)$ is needed to consider the case that the stress is the state variable. The strain as a function of stress is written in terms of the product log function, which is the inverse of the function $z = x \exp(x)$. The product log of z , which is denoted $h(z) = x$, exists because $x \exp(x)$ is monotonically increasing, but $h(z)$ must be computed numerically because it has no closed form. Then the strain is written in terms of the stress,

$$E = \frac{1}{\sqrt{2c_1}} \sqrt{h(S^2/a)}, \quad (22)$$

where $a = 2c^2c_1$.

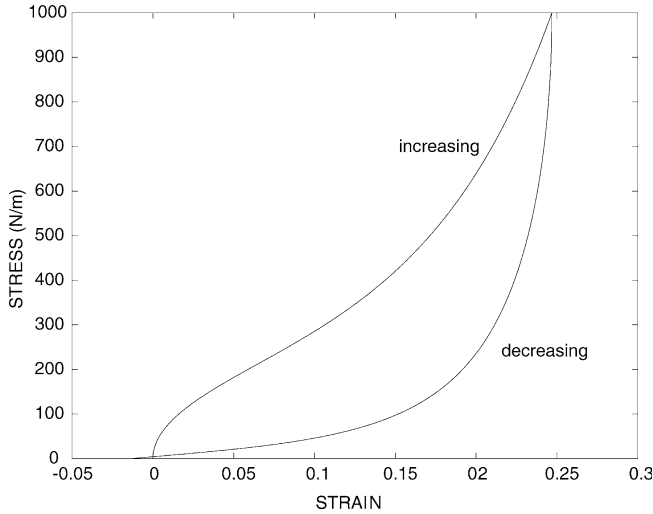


Fig. 3 Prediction of the uniaxial stress–strain curve controlled by a monotonically increasing stress. The long-term Fung exponential model has constants $c = 0.88$ and $c_1 = 13$. The *upper curve* is the loading at a stress rate of 0.05 Pa/s for 20,000 s, and the *lower curve* is the unloading at the same rate. The relaxation modulus is $k_E = 0.00001$. The solution is by MATLAB ode23

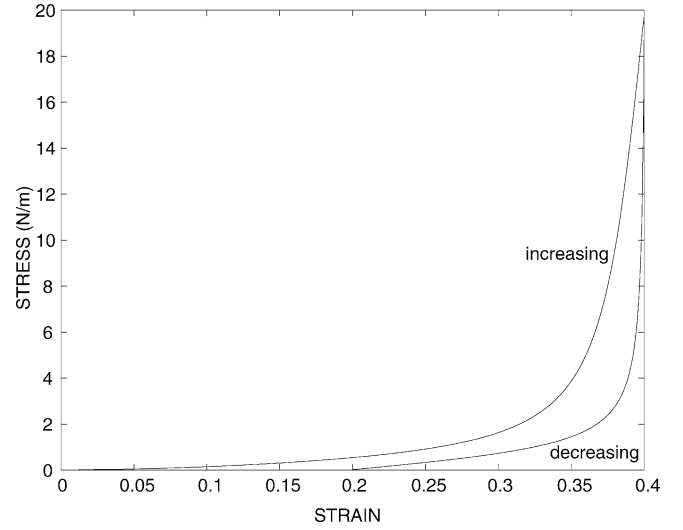


Fig. 4 Prediction of the uniaxial stress–strain curve controlled by a monotonically increasing strain. The long-term Fung exponential model has constants $c = 0.88$ and $c_1 = 13$. The *upper curve* is the loading at a strain rate of 0.0002/s for 2,000 s, and the *lower curve* is the unloading at the same rate. The relaxation modulus is $k_E = 0.0001$. The solution is by MATLAB ode23

The evolution (Eq. 6) for stress as a state variable and strain as a control variable is

$$\dot{S} = -k_S \left(\frac{2c_1 S^2 (1 + h(S^2/a))^2}{h(S^2/a)} \right) \left[\frac{1}{\sqrt{2c_1}} \sqrt{h(S^2/a)} - E(t) \right]. \quad (23)$$

The stress–strain curve (Fig. 4) obtained from the evolution equation 23 with $k_S = 0.0001$ in strain control for $E(t) = 0.002t$ captures concave up behavior as well as loading–unloading hysteresis.

3.3 Skalak model

The Skalak et al. (1975) model has strain energy, which is here taken as the long-term manifold relation,

$$\phi(\lambda) = c \left[2(\lambda^4 - 2\lambda^2 + 1) + \Gamma(\lambda^4 - 1)^4 \right], \quad (24)$$

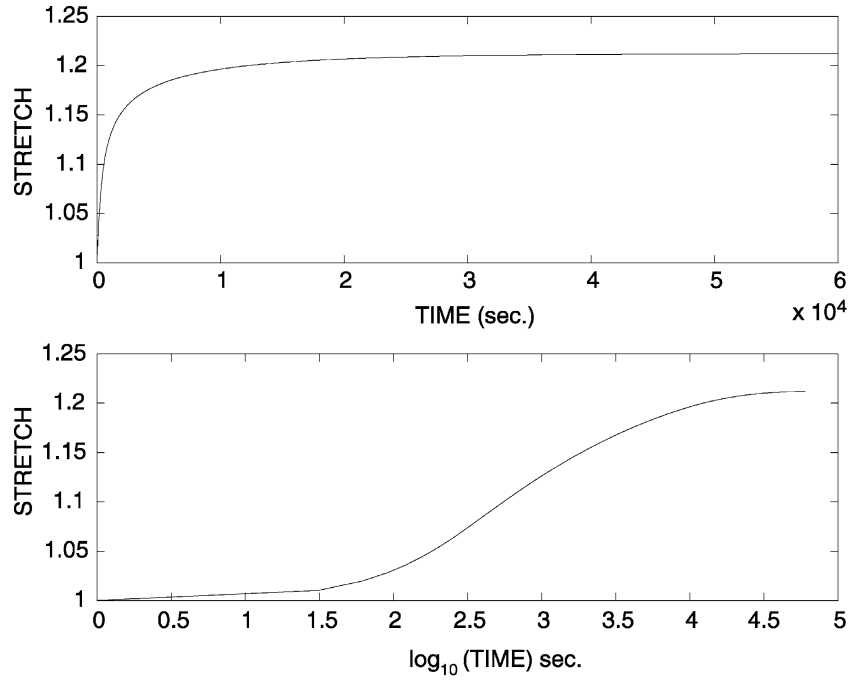
where $c > 0$ has units of force per length and $\Gamma > 0$ is dimensionless.

The first Piola–Kirchhoff stress, P , versus stretch relation is given by

$$P = \frac{d\phi}{d\lambda} = c \left[2(-4\lambda + 4\lambda^3) + 16\Gamma\lambda^3(-1 + \lambda^4)^3 \right],$$

whose graph has the same qualitative shape as that of the Fung models. Increasing Γ moves the curve slightly to the left and increases the magnitude for a given stretch. The evolution (Eq. 6) for λ the state and P the control variables is

Fig. 5 Creep response if the long-term manifold is represented by a Skalak model, under a stress of $P=9$, with $\lambda(0)=1$, $c=1$, $\Gamma=0.1$, and $k=0.01$. **a** Creep stretch versus time. **b** creep stretch versus $\log_{10}(\text{time})$. The solution is by MATLAB ode23



$$\begin{aligned} \dot{\lambda} &= k \left(\frac{d^2 \phi}{d\lambda^2} \right)^{-2} \left[\frac{d\phi}{d\lambda} - P \right] \\ &= (k/c^2) [2(-4 + 12\lambda^2) + 192\Gamma\lambda^6(-1 + \lambda^4)^2 \\ &\quad + 48\Gamma\lambda^2(-1 + \lambda^4)^3]^{-2} \\ &\quad \times c \left[2(-4\lambda + 4\lambda^3) + 16\Gamma\lambda^3(-1 + \lambda^4)^3 - \frac{P}{c} \right] \end{aligned} \quad (25)$$

The creep behavior predicted by Eq. 25 is shown in Fig. 5 for $\lambda(0)=1$, $\Gamma=0.1$, $k=0.01$, $c=1$, and $P=9$. This model reproduces the qualitative shape of the experimental creep data given by Tanaka and Fung (1974) because the stretch versus $\log_{10}(\text{time})$ curve (Fig. 5b) is S-shaped. If k is made larger, say $k=1$, then the stretch versus $\log_{10}(\text{time})$ curve is nearly a straight line until the long-term manifold is reached; at that point the stretch oscillates about its long-term value. Therefore, in this case the transient response predicted by the evolution equation with Skalak long-term model is nearly logarithmic.

A stress relaxation model is mathematically harder to obtain because the long-term stretch must be written as a function of the stretch. A closed form seems impossible.

3.4 Behavior under sinusoidal loading

Periodic, or nearly periodic, loadings of biological soft tissue are common in vivo. The simplest of these loads is harmonic. Some experiments have been reported for harmonic loading in which dynamic moduli are obtained under the assumption that the biological tissue is linearly viscoelastic. Typically such tests are performed under a

very small forcing amplitude with the consequence that the response to sinusoidal forcing often appears harmonic as predicted by a linear viscoelastic model. However, one cannot deduce from an apparently harmonic response that the material is necessarily linear viscoelastic. The measured phase shift between the forcing function and the response is a constant and a material property only for linear viscoelastic materials. Of more interest for soft tissue is the possibly nonharmonic response to larger amplitude harmonic forcing in which the nonlinear behavior of the system becomes apparent.

Cyclic loading experiments were performed by Apter and Marquez (1968) on aortic tissue containing muscle as well as elastin and collagen, in circumferential strain control at small strain amplitudes and various mean strains. Their rudimentary apparatus, which yielded numerical values as estimates from the graph of an x - y pen-plotter, applied an amplitude of about 1% strain at frequencies between 0.01 and 21 Hz. In the range up to 1 Hz, the response to a sinusoidal stretch was observed to be sinusoidal. The force amplitude of the response increased with higher frequencies. The behavior was similar to that of a standard linear viscoelastic solid only at small frequencies, with the stress response lagging the forcing strain on a time-dependent plot. Apter and Marquez observed a 90° phase shift of the response at about 5 Hz. A 90° phase shift is taken by them as the definition of resonance frequency, where the magnitude of the dynamic modulus, $|G(\omega)|$, is approximately the largest. This definition only makes sense for linear viscoelasticity. Apter and Marquez reported that adventitial collagen exhibited no stress response phase shift under sinusoidal strains, an observation which they attributed to elastic behavior. However, their medial

collagen response could be fit by a Kelvin–Voigt model because they observed little stress relaxation of the collagen.

The strain controlled experiments of Patel et al. (1970) measured the properties of a canine aorta in the radial, but not in the longitudinal or circumferential directions. A small amplitude sinusoidal strain was superimposed on a compressive mean strain of about -0.5 with frequency fixed between 2 and 40 Hz. When the stress response was not harmonic, only the fundamental harmonic was used in analysis. The data nearly fit a parallel spring and dashpot Kelvin–Voigt model in the frequency range of 2–40 Hz but not well in the range of 0–2 Hz. They estimated, from a spring and dashpot model, a 200 Hz resonance in contrast to the 5–8 Hz resonance in the Apter and Marquez (1968) data. They found that their linear model held if the strain amplitude superposed on a given strain was held below 4%.

Gow and Taylor (1968) measured the dynamic moduli of the arteries in living dogs. They observed, contrary to their expectations, in some cases when measuring the response of the diameter of the artery to a sinusoidal pressure load, that the diameter actually led the pressure when only the first harmonics of the response were considered. They attributed this behavior to nonlinearities in the pressure–diameter relationship. These nonlinearities affect the higher harmonics in a Fourier analysis of the pressure and diameter waves.

The model of Eq. 6 is applied to the Tanaka–Fung model for the long-term manifold, again using Fung’s notation. In stress control,

$$\begin{aligned} \lambda = & -k[\alpha G(\infty)(T^* + \beta) \exp(\alpha(\lambda - \lambda^*))]^{-2} \\ & \times \{-[\sigma_o + \sigma_a \sin(\omega t)] + G(\infty) \\ & \times [(T^* + \beta) \exp(\alpha(\lambda - \lambda^*) - \beta)]\} \end{aligned} \quad (26)$$

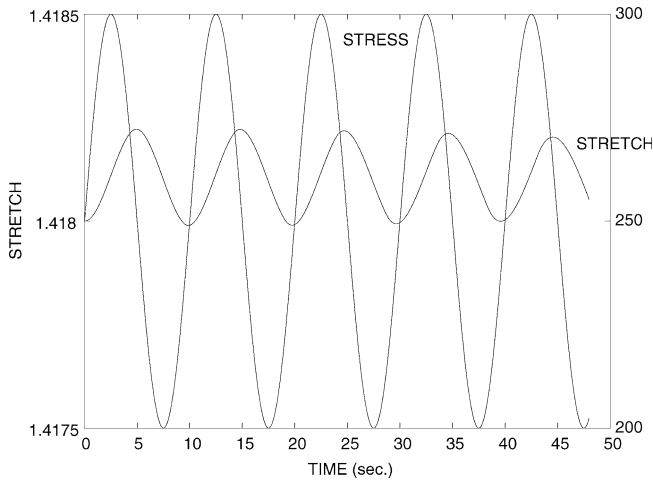


Fig. 6 Prediction of the stretch response to a sinusoidal stress $T = 250 + 50\sin(\omega t)$ g/cm² for a long-term Tanaka–Fung model with constants $\lambda^* = 1.6$, $T^* = 535$ g/cm², $\alpha = 0.57$, $\beta = 910/\alpha$ g/cm², and $G(\infty) = 0.7692$. Also $k = 1.0$, $\omega = 0.1(2\pi)$, and $\lambda(0) = 1.418$. The curve with the larger amplitude is the applied stress. The solution is by MATLAB ode23

Figure 6 shows the response for $k = 1.0$ to a sinusoidal stress with a forcing frequency of 0.1 Hz, the low end of the Apter and Marquez experiments. The initial stretch is taken to be the value corresponding on the long-term manifold to the mean value of the applied stress. The stretch lags the stress. The response is similar to a harmonic one, but the lag between stress and strain decreases from cycle to subsequent cycle. In a linear viscoelastic model, the lag is constant. Further, the amplitude of the response decreases on subsequent cycles. As k is increased, the magnitude of the response increases.

An increase in the forcing frequency, ω , causes the stretch response to undergo a transition from a qualitatively harmonic shape. Such behavior is common in nonlinear dynamic models. The clearly nonlinear response is shown in Fig. 7 for 1 Hz forcing.

In stretch control,

$$\begin{aligned} T = & -k_T[\alpha(T + \beta G(\infty))]^2 \left\{ \lambda^* + \frac{1}{\alpha} \left[\ln \left(\frac{[T/G(\infty)] + \beta}{T^* + \beta} \right) \right] \right. \\ & \left. - [\lambda_0 + \lambda_a \sin(\omega t)] \right\}, \end{aligned} \quad (27)$$

where the initial stress corresponds to λ_o and the cyclic amplitude is λ_a .

4 Temperature dependence in uniaxial loading

Biological soft tissue normally operates in a narrow temperature range. The construction of this paper can represent the evolution of temperature. Soft biological tissues contract when heated (Roy 1880), just as rubber

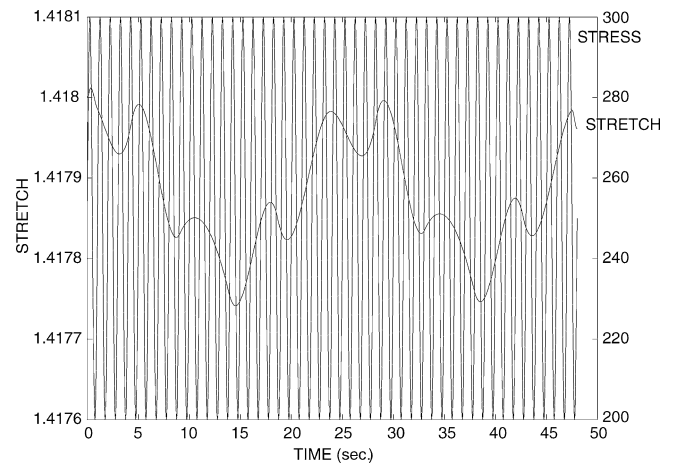


Fig. 7 Prediction of the stretch response to a sinusoidal stress $T = 250 + 50\sin(\omega t)$ g/cm² for a long-term Tanaka–Fung model with constants $\lambda^* = 1.6$, $T^* = 535$ g/cm², $\alpha = 0.57$, $\beta = 910/\alpha$ g/cm², and $G(\infty) = 0.7692$. Also $k = 1.0$, $\omega = 2\pi$, and $\lambda(0) = 1.418$. The curve with the larger amplitude is the applied stress. The solution is by MATLAB ode23

does, in contrast to metals. Also soft tissue temperature increases slightly when tissue is stretched, as does rubber. These facts suggested to Fung (1972) the possibility of applying the thermodynamics of entropic elasticity taken from classical rubber theory (see also Fung 1993, Chapter 7.4). The entropic theory postulates that elastic loading may be resolved into two components, one due to the entropy and the other to the internal energy (Anthony et al. 1942). This idea seems to have been based on an analogy with the van der Waals gas model. The component due to internal energy was claimed to be insignificant. Experimental data on the temperature dependence of stresses is collected for this theory by measuring the stress on a specimen held at constant length and volume as the temperature is varied. Such a test on rubber produces a nearly linear stress–temperature relation. Apparently, little data describing the temperature dependence of stress in soft tissue exists in the literature. Rigby (1979) reported that a rat-tail tendon in a 0.9% NaCl solution the force decreased only slightly as temperature was increased at 0.5° per min from 20 to 60°C. At about 60°C, the shrinkage temperature of collagen, it exhibited a sharp increase. Unfortunately, the entropic theory is controversial even for rubber elasticity (Rosa et al. 1992). Fung (1993, p. 269) points out that the entropic theory does not work if the material structure changes with temperature, as is the case for biological materials such as elastin, which adsorbs moisture as temperature changes.

For the purpose of demonstrating the construction of an evolution model, which represents the evolution of the two state variables, stretch and temperature, assume that soft biological tissue has a long-term energy function which depends on temperature, θ_c , measured from a reference state and the Green strain, $E = 0.5(\lambda^2 - 1)$, where λ is the uniaxial stretch.

$$\psi(E, \theta_c) = c\theta_c \{ \exp[c_2(\theta_c)E^2] - 1 \} + F(\theta_c). \quad (28)$$

The function $F(\theta_c)$, which ensures the proper entropy expression, must be determined from additional experimental data. Then $d^2F/d\theta_c^2$ is determined from the relation defining the specific heat at constant deformation, $c_L(\theta) = -\theta (\partial^2 \psi / \partial \theta^2)$. This information for soft tissue is apparently not yet available in the literature. Therefore, a linear expression will be taken that is similar to rubber for this illustration; $c_L(\theta) = 964.4 + 1.59\theta$. Then $F(\theta_c) = -964.4[\theta \ln(\theta) - \theta] - 0.5(1.59)\theta^2 + C\theta + D$. The constants $c = 0.88/25$ and $c_2 = 26$ are based on those of Shah and Humphrey (1999), where c is adjusted for the reference temperature of 25°C, and c_2 is assumed independent of temperature.

This Helmholtz function implies that the generalized energy, Ψ^* , is

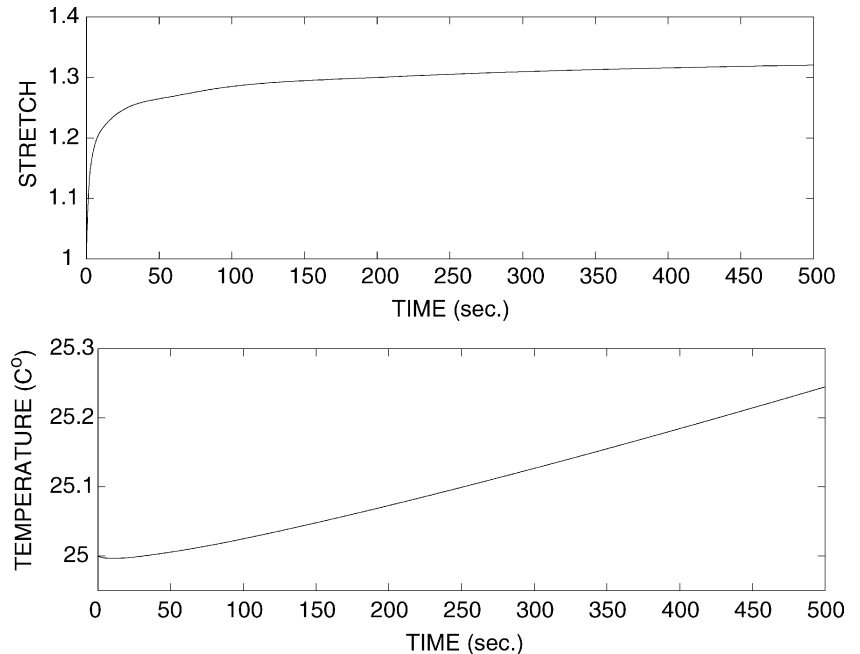
$$\psi^*(E, \theta; S, \eta) = c\theta \exp(c_2 E^2) - 964.4[\theta \ln(\theta) - \theta] - 0.5(1.59)\theta^2 + C\theta + D - SE - \theta\eta. \quad (29)$$

If the second Piola–Kirchhoff stress S and the entropy are control variables, then the isentropic temperature evolution is coupled with the creep strain response by Eq. 6 which becomes

$$\begin{bmatrix} \frac{dE}{dt} \\ \frac{d\theta}{dt} \end{bmatrix} = k \begin{bmatrix} \frac{\partial^2 \psi}{\partial E^2} & \frac{\partial^2 \psi}{\partial E \partial \theta} \\ \frac{\partial^2 \psi}{\partial \theta \partial E} & \frac{\partial^2 \psi}{\partial \theta^2} \end{bmatrix}^{-2} \begin{bmatrix} S & -\partial \psi / \partial E \\ -\tilde{\eta} & -\frac{\partial \psi}{\partial \theta} \end{bmatrix}, \quad (30)$$

where S and η are the sudden stress perturbation and the sudden entropy perturbation, respectively, which are held fixed during the process, and where $\tilde{\eta} = \eta - C$. The evolution of the stretch is obtained by setting $dE/dt = \lambda$

Fig. 8 Temperature change during uniaxial loading of a material with a Fung type long-term model with $c = 0.88/25$ and $c_2 = 26$. **a** The evolution of the stretch under a constant stress of 22,000 Pa. **b** The time-dependent increase in temperature. The initial conditions are $\lambda(0) = 1$ and $\theta(0) = 25^\circ\text{C}$. The solution is by MATLAB ode15s



($d\lambda/dt$) in Eq. 30. The evolution of stretch and temperature is shown in Fig. 8 for $S=22,000$ Pa and $\tilde{\eta}=3,300$. The stretch asymptotically approaches the equilibrium value and the temperature increases about 0.3°C for $k=0.01$ in 500 s. Larger values of k make the stretch approach its asymptotic value more quickly and increase the temperature rise. There is a slight initial decrease in temperature after the load is applied; this dip is exaggerated at smaller values of k . The temperature increase predicted by this model during an axial loading is qualitatively the same as that found experimentally for biological soft tissue.

5 Evolution equations based on the Holzapfel et al. long-term three-dimensional model for healthy artery tissue

The construction of the evolution equations can be applied to represent the time-dependent behavior when the elastic Holzapfel et al. (2000) composite three-dimensional energy function for healthy artery tissue is the model for the long-term behavior. This model assumes a two-layer tissue with reinforcing collagen fibrils. Tensors $\mathbf{A}_1$ and $\mathbf{A}_2$ define the orientation of the two families of parallel collagen fibers in each layer. In cylindrical coordinates (r, θ, z) , the tensors describing the collagen fibers are defined by $\mathbf{A}_i = \mathbf{a}_i \otimes \mathbf{a}_i$, where

$$\mathbf{a}_1 = (0, \cos \beta, \sin \beta) \text{ and } \mathbf{a}_2 = (0, \cos \beta, -\sin \beta).$$

This tensor implies that any small components of the fibers in the radial direction are neglected. The angle β is on the order of 29° for the media; it is about 62° in the adventitia.

Holzapfel et al. (2000) work in terms of isochoric stresses and strains. If the deformation gradient tensor $\mathbf{F} = J\tilde{\mathbf{F}}$, where $J = \det(\mathbf{F})$, then by definition $\tilde{\mathbf{C}} = \tilde{\mathbf{F}}^t \mathbf{F}$. Their long-term anisotropic elastic energy for a two-layer tissue is $\Psi = \psi_M + \psi_A$, where the functions ψ_M and ψ_A represent the strain energy in the media and adventitia layers, respectively. These functions have the form

$$\psi = \frac{c}{2}(\bar{I}_1 - 3) + \frac{k_1}{2k_2} \sum_{i=4,6} \left\{ \exp[k_2(\bar{I}_i - 1)^2] - 1 \right\}, \quad (31)$$

where $\bar{I}_1 = \text{Tr}(\tilde{\mathbf{C}})$, $\bar{I}_4 = \tilde{\mathbf{C}} : \mathbf{A}_1$, $\bar{I}_6 = \tilde{\mathbf{C}} : \mathbf{A}_2$, and k_1 , k_2 , and c are the material constants. The first term, a neo-Hookean model, accounts for the isotropic matrix, which is mostly elastin and muscle cells, and the exponential terms account for the anisotropic collagen reinforcing fibers.

Assume $\det(\mathbf{F})=1$. If $\mathbf{F}$ is diagonal, then $\bar{I}_4 = \tilde{\mathbf{C}} : \mathbf{A}_1 = a_{11}\lambda_1^2 + a_{22}\lambda_2^2 + a_{33}\lambda_3^2$, where $a_{11}=0$, $a_{22}=\cos^2\beta$ and $a_{33}=\sin^2\beta$. Also $\bar{I}_6 = \tilde{\mathbf{C}} : \mathbf{A}_2 = \bar{I}_4$ in this case. Then, for example, $\bar{I}_4 - 1 = \cos^2(\beta)\lambda_2^2 + \sin^2(\beta)\lambda_3^2 - (\sin^2\beta + \cos^2\beta) = 2\cos^2(\beta)E_2 + 2\sin^2(\beta)E_3$. The exponential term in the Holzapfel et al. model is

$$\exp[k_2(\bar{I}_4 - 1)^2] = \exp[k_2(4\cos^4(\beta)E_2^2 + 4\sin^4(\beta)E_3^2 + 8\cos^2(\beta)\sin^2(\beta)E_2E_3)]. \quad (32)$$

The term $\exp[k_2(\bar{I}_6 - 1)^2]$ is identical. Therefore, in the case that $\mathbf{F}$ is diagonal, the Holzapfel et al. model is of the same form as, and is essentially a three-dimensional version of, the Fung model, except that the material constants are given a physical interpretation.

The evolution equations 6 do not maintain incompressibility during a deformation. This behavior is similar to the plastic deformation of metals, for which experiments show that the volume is not preserved during plastic deformation, although it is preserved from before the load is applied to after the load is removed. These evolution equations define a material which is incompressible in the long term.

Definition 5.1. A material is said to be long-term incompressible under a load if its long-term behavior is described by an incompressible energy function in the sense that $\lambda_1\lambda_2\lambda_3=1$ in the long term.

The calculations here are based on the principal stretches and are presented in terms of the conjugate pair composed of the Green tensor $\mathbf{E}$ and the second Piola–Kirchhoff stress, $\mathbf{S}$. The evolution equations 6 are, for stress control,

$$\dot{\mathbf{E}} = -k \left(\frac{\partial^2 \Psi}{\partial \mathbf{E}^2} \right)^{-2} \left[-\mathbf{S}(t) + \frac{\partial \Psi}{\partial \mathbf{E}} \right]. \quad (33)$$

By the discussion in the Appendix, Eq. 33 becomes, in the case that $\mathbf{F}$ is diagonal,

$$\left[\frac{dE_2}{dt} \right] = k \left[\begin{array}{cc} \frac{\partial^2 \Psi}{\partial E_2^2} & \frac{\partial^2 \Psi}{\partial E_2 \partial E_3} \\ \frac{\partial^2 \Psi}{\partial E_3 \partial E_2} & \frac{\partial^2 \Psi}{\partial E_3^2} \end{array} \right]^{-2} \left[\begin{array}{c} S_2 - \frac{\partial \Psi}{\partial E_2} \\ S_3 - \frac{\partial \Psi}{\partial E_3} \end{array} \right], \quad (34)$$

where $\dot{E}_i = \lambda_i \dot{\lambda}_i$ and $\Psi = \psi_M + \psi_A$ for which ψ_M and ψ_A have the form

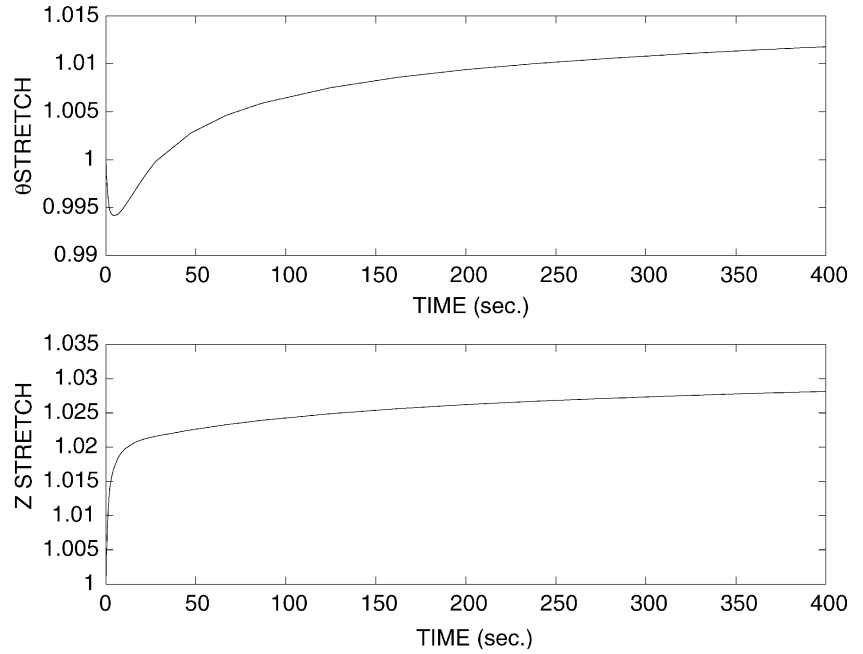
$$\begin{aligned} \psi(E_2, E_3) &= \frac{c}{2}(\bar{I}_1 - 3) + \frac{k_1}{2k_2} \sum_{i=4,6} \left\{ \exp[k_2(\bar{I}_i - 1)^2] - 1 \right\} \\ &= c(E_2 + E_3) + \frac{k_1}{k_2} \left\{ \exp[c_1 E_2^2 + c_2 E_3^2 + c_3 E_2 E_3] - 1 \right\}, \end{aligned} \quad (35)$$

and $c_1 = 4k_2 \cos^4\beta$, $c_2 = 4k_2 \sin^4\beta$, and $c_3 = 8k_2 \cos^2\beta \sin^2\beta$. The 2 in front of the exponential in the second line accounts for both the terms involving $(\bar{I}_4 - 1)$ and $(\bar{I}_6 - 1)$. The derivatives are

$$\begin{aligned} \frac{\partial \psi}{\partial E_2} &= c + \frac{k_1}{k_2} [2c_1 E_2 + c_3 E_3] \exp[c_1 E_2^2 + c_2 E_3^2 + c_3 E_2 E_3]; \\ \frac{\partial \psi}{\partial E_3} &= c + \frac{k_1}{k_2} [2c_2 E_3 + c_3 E_2] \exp[c_1 E_2^2 + c_2 E_3^2 + c_3 E_2 E_3]; \\ \frac{\partial^2 \psi}{\partial E_2^2} &= \frac{k_1}{k_2} [2c_1 + (2c_1 E_2 + c_3 E_3)^2] \exp[c_1 E_2^2 + c_2 E_3^2 + c_3 E_2 E_3]; \\ \frac{\partial^2 \psi}{\partial E_3^2} &= \frac{k_1}{k_2} [2c_2 + (2c_2 E_3 + c_3 E_2)^2] \exp[c_1 E_2^2 + c_2 E_3^2 + c_3 E_2 E_3]; \\ \frac{\partial^2 \psi}{\partial E_2 \partial E_3} &= \frac{k_1}{k_2} [c_3 + (2c_1 E_2 + c_3 E_3)(2c_2 E_3 + c_3 E_2)] \\ &\quad \times \exp[c_1 E_2^2 + c_2 E_3^2 + c_3 E_2 E_3]. \end{aligned}$$

Experimenters sometimes excise tissue and apply biaxial loads to a flat specimen. Creep of a two-layer

Fig. 9 Creep of a two-layer flat specimen cut along the θ and z directions under equibiaxial stress of 22,000 Pa. The initial conditions are $\lambda_\theta(0) = \lambda_z(0) = 1$, and $k = 5$. The solution is by MATLAB ode15s



specimen cut along the θ and z directions under biaxial dead loads causing fixed stresses, S_θ and S_z , in the respective principal directions is described by the evolution model, Eq. 6. Choose a coordinate system in which the stretches are principal, and assume the isothermal Holzapfel et al. two-layer model for the long-term behavior. As an example, the constants given for rabbit carotid tissue by Holzapfel et al. (2000, p. 36) are applied for an equibiaxial stress of 22,000 Pa. The transient response of the stretches, λ_θ and λ_z , is shown in Fig. 9. The specimen is stiffer in the θ -direction. The calculations were made in MATLAB using the stiff integrator, ode15s.

The evolution equations 34 differ from those presented in Holzapfel et al. (2002) where the Holzapfel et al. (2000) model was used for the long-term behavior term in the Bonet construction that adopts Maxwell models to represent dissipation. The Bonet–Holzapfel viscoelastic model assumes a splitting of the stress into a sum of the long-term stress and a transient (viscoelastic) stress which tends to zero in the long term. Apparently, a generalized Maxwell evolution model was assumed because in that model, as opposed to other simple spring and dashpot models like the Kelvin–Voigt, the stress relaxes to zero under a constant strain. It, therefore satisfies the splitting of the stress into a sum of the long-term stress and a transient (viscoelastic) stress which tends to zero in the long term.

The Holzapfel et al. (2002) model assumes an energy function of the form

$$\Psi = U(\mathbf{X}, J) + \bar{\Psi}(\mathbf{X}, \bar{\mathbf{C}}, \mathbf{A}_1, \mathbf{A}_2) + \sum_{\alpha} \Psi_{\alpha}(\mathbf{X}, \bar{\mathbf{C}}, \mathbf{A}_1, \mathbf{A}_2, \Gamma_{\alpha}), \quad (36)$$

where Γ_{α} are viscous variables and $\mathbf{X}$ denotes the point in the body. For computational reasons, they allow the

tissue to be slightly compressible, as accounted for in the term, $U(\mathbf{X}, J)$.

The assumed dissipation is given by the set of linear differential equations

$$\dot{\mathbf{Q}} + \frac{\mathbf{Q}}{\tau} = \beta^{\infty} \dot{\mathbf{S}}_{iso},$$

for constants τ and β^{∞} , that are valid for a time interval $0 < t \leq T$ and for small perturbations from equilibrium. Here, $\mathbf{Q}$ is the non-equilibrium viscoelastic contribution to the stress and is defined from the energy function as

$$\mathbf{Q}_{\alpha} = 2 \frac{\Psi_{\alpha}(\mathbf{X}, \bar{\mathbf{C}}, \mathbf{A}_1, \mathbf{A}_2, \Gamma_{\alpha})}{\partial \mathbf{C}}.$$

The evolution equations in Holzapfel et al. (2002) are linear involving two new coefficients, τ and β^{∞} for each $\mathbf{Q}_{\alpha}$, in contrast to the single value, k , in Eq. 6. The nonlinear evolution Eq. 6 proposed here are not restricted to small perturbations from equilibrium.

6 Viscoelastic aneurysm model

This viscoelastic construction is applied to the Shah and Humphrey (1999) elastodynamical model for a saccular aneurysm as an illustration. Shah and Humphrey (1999) represented a thin-necked aneurysm as a thin-walled, isotropic, nonlinear elastic sphere without a hole which is surrounded by the viscous, Newtonian, cranial spinal fluid and forced by a known intra-saccular pressure. The cranial spinal fluid is neglected in this illustration.

The sphere is in load control subjected to a transmural radial pressure, $P(t)$. The sphere is assumed to deform uniformly as measured by a single state variable,

the in-plane membrane stretch $\lambda(t) = 2\pi a(t)/2\pi A = a(t)/A$, where $a(t)$ is the current deformed radius and A is the original undeformed radius. Here ρ denotes the constant mass density of the membrane, and H the undeformed thickness of the membrane. The incompressibility constraint on the membrane volume requires that the deformed thickness of the membrane is $h = H/\lambda^2$ because the volume is preserved, $4\pi A^2 H = 4\pi a^2 h$. According to Shah and Humphrey, this problem can be assumed to be two-dimensional with deformation gradient $\mathbf{F}$ having diagonal entries equal to λ .

The control variable, the second Piola–Kirchhoff stress $\mathbf{S}$, is related to the transmural pressure and the stretch acceleration by the equation of motion in the r -direction. The Shah and Humphrey (1999, Eq. 4) is

$$P(t) - \frac{2T}{A\lambda} = \rho A h \ddot{\lambda}, \quad (37)$$

where T is the Cauchy stress. The second Piola–Kirchhoff stress is related to the Cauchy stress by $\mathbf{S} = \mathbf{J}\mathbf{F}^{-1}\mathbf{T}\mathbf{F}^{-t}$. In this two-dimensional model, $J = \lambda^2$ so that $T = S$. Therefore

$$S = \frac{A\lambda}{2} \left[P(t) - \rho A h \ddot{\lambda} \right] = \frac{A\lambda}{2} \left[P(t) - \frac{\rho A H}{\lambda^2} \ddot{\lambda} \right]. \quad (38)$$

The energy function representing the long-term behavior is

$$\Psi(E_2, E_3) = c \{ \exp[c_1 E_2^2 + c_2 E_3^2 + c_3 E_2 E_3] - 1 \}. \quad (39)$$

But $E_2 = E_3 = 0.5(\lambda^2 - 1)$. So

$$\psi = c [\exp(c_t E_2^2) - 1]. \quad (40)$$

where $c_t = c_1 + c_2 + c_3$.

$$\frac{\partial \psi}{\partial E_2} = 2cc_t E_2 [\exp(c_t E_2^2)]$$

$$\frac{\partial^2 \psi}{\partial E_2^2} = 2cc_t (1 + 2cc_t E_2^2) [\exp(c_t E_2^2)]$$

The evolution equation 6 becomes

$$\dot{E}_2 = k \left(\frac{\partial^2 \psi}{\partial E_2^2} \right)^{-2} \left(S - \frac{\partial \psi}{\partial E_2} \right). \quad (41)$$

Using $\dot{E}_2 = \lambda \dot{\lambda}$, writing the equation in terms of λ and rearranging yields the governing differential equation,

$$\begin{aligned} \ddot{\lambda} &= \frac{2\dot{\lambda}}{\rho A^2 H} \left[-\frac{\dot{\lambda}}{k} \left(\frac{\partial^2 \psi}{\partial E_2^2} \right)^2 + \frac{A\dot{\lambda}}{2} P(t) - \frac{\partial \psi}{\partial E_2} \right] \\ &= \frac{2\dot{\lambda}}{\rho A^2 H} \left[-\frac{\dot{\lambda}}{k} \left[2cc_t (1 + 0.5cc_t (\lambda^2 - 1)^2) \exp[0.25c_t (\lambda^2 - 1)^2] \right]^2 \right. \\ &\quad \left. + \frac{A\dot{\lambda}}{2} P(t) - cc_t (\lambda^2 - 1) \exp[0.25c_t (\lambda^2 - 1)^2] \right]. \end{aligned}$$

Shah and Humphrey (1999) assume the values $A = 0.003$ m, $H = 0.000278$ m, $\rho = 1,050$ kg/m³, $c = 0.88$ and $c_t = 13$ obtained from the Kyriacou and Humphrey (1996) numerical fit to the experimental data of Scott

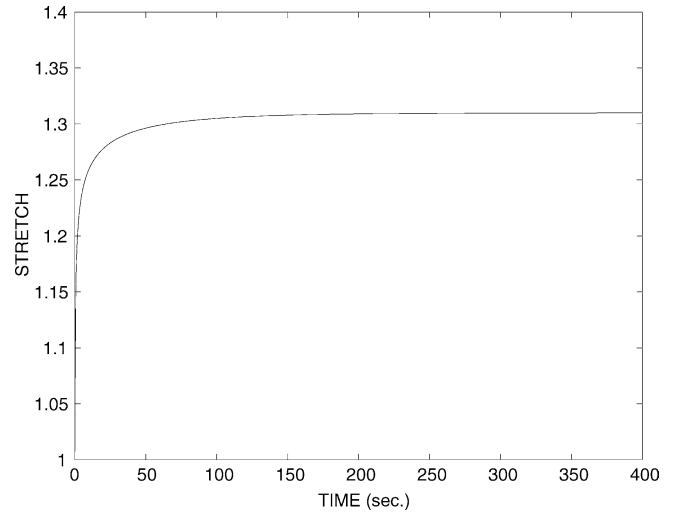


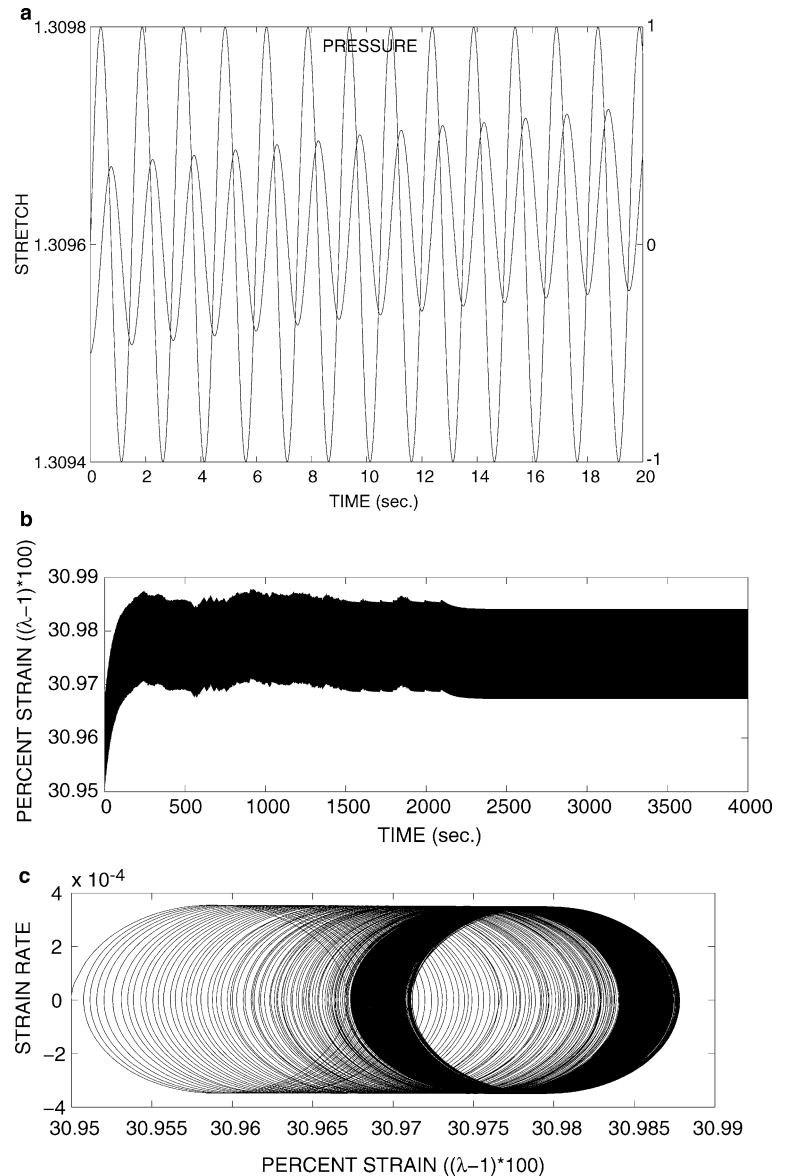
Fig. 10 The viscoelastic creep behavior of the modified Shah–Humphrey saccular aneurysm model as the internal pressure is suddenly increased from zero to 22,000 Pa with initial conditions $\lambda = 1$ and $\dot{\lambda}(0) = 0$ and $k = 15$. The MATLAB stiff integrator ode15s was used to solve the evolution equation

et al. (1972), and these values are used in the following two computations for creep and sinusoidal loading.

If the internal pressure in an undeformed saccular aneurysm is suddenly increased from zero to 22,000 Pa and if $\dot{\lambda}(0) = 0$ and $k = 15$, then the stretch increases asymptotically from 1 to nearly $\lambda = 1.31$, the long-term stretch at this load (Fig. 10). The MATLAB stiff integrator ode15s was used to solve the evolution equation.

The evolution construction is applied to a sinusoidal pressure loading $22,000 + 7,000\sin(\omega t)$, where $\omega = 2\pi/1.5$ to represent the blood pressure. The initial conditions in Fig. 11 were $\lambda(0) = 1.3095$ and $\dot{\lambda}(0) = 0$, with $k = 15$. Figure 11a for a short portion of the evolution time shows that the stretch lags the applied sinusoidal pressure load, as expected. The lag is initially about $\pi/4$ but increases over time because of the nonlinear viscoelastic behavior. In this example, the rate of change of the stretch, $\dot{\lambda}$, is very small. The MATLAB stiff integrator ode15s was used to solve the evolution equation. Figure 11b shows the full evolution to a limit cycle, and Fig. 11c is a phase diagram showing the limit cycle as the mean amplitude grows slowly to the asymptotic limit. The percent engineering strain is plotted in Fig. 11b and c because Matlab prints at most five significant figures on the axes of graphs. The evolution equation was also solved for values of k up to 500. The larger that k is, the larger the amplitude of the limit cycle. The stretch rate increases somewhat with an increase in k . At smaller k , the mean continues to grow approaching the asymptote for some time. The response to cyclic loading is expected to approach a stable cycle because several experimenters have conditioned soft tissue specimens by cycling the control load until a stable response is observed before taking measurements.

Fig. 11 Stretch response of the modified Shah–Humphrey saccular aneurysm model under a sinusoidal pressure loading $22,000 + 7,000\sin(\omega t)$, where $\omega = 2\pi/1.5$. The initial conditions are $\lambda(0) = 1.3095$ and $\dot{\lambda}(0) = 0$, with $k = 15$. The MATLAB stiff integrator ode15s was used to solve the evolution equation. **a** The large amplitude curve is the scaled pressure to exhibit the stretch lag behind the applied pressure in a short range of times. **b** The long-term stretch response showing the approach to a limit cycle. **c** The phase diagram for the response in **b**



7 Conclusion

The proposed nonlinear viscoelastic model for solids (Eq. 6) represents all types of time-dependent behavior. Because a unique long-term manifold is assumed to exist for both stress and strain controlled relaxation (stress relaxation and creep), the model can represent any process which is either stress or strain controlled. The only experimental data needed for this model is the time-independent long-term model and the relaxation constant, k , for the state variable in question. There is no need to fit time-dependent empirical functions to the data as required in many existing nonlinear viscoelastic models. Such fits are notoriously difficult to achieve. This method of constructing a nonlinear viscoelastic model applies to any biological material for which data and a model for the long-term behavior and the relax-

ation constant may be obtained. This construction fits the modeling of biological tissue into the same framework as those previously constructed for the viscoelastic behavior of synthetic polymers and for the viscoplastic behavior of metals.

The evolution model qualitatively represents the creep data which Tanaka and Fung could not fit with their model and load–unload hysteresis in monotonic loading. The effect of variation of the relaxation modulus, k , has been exhibited in several numerical simulations. The role of the amplitude and the frequency of harmonic loading can be estimated using the evolution model. Small amplitudes and frequencies produce a nearly harmonic response as if the behavior were similar to linear viscoelastic. Application of the evolution construction has been shown to several important two-dimensional problems. In any case, the point of this

paper has been to illustrate the construction of the model rather than to fit a particular set of data.

Appendix

Evolution equation when the strain energy is a function of a tensor

Let $\mathbf{E}$ be a second order tensor of state variables, and let $\mathbf{H}$ be the second order tensor of conjugate control variables. Assume that the energy function has the form

$$\psi(\mathbf{E}; \mathbf{H}) = \phi(\mathbf{E}) + \mathbf{E} : \mathbf{H}.$$

The affinities are the second order tensor, $\mathbf{X} = \partial\psi/\partial\mathbf{E}$. The affinities are a function of $\mathbf{E}$, so that $\mathbf{X} = h(\mathbf{E})$. Assume that h is invertible so that $\psi(\mathbf{E}; \mathbf{H})$ can be written in terms of the affinities for fixed control variables, $\bar{\psi}(\mathbf{X}; \mathbf{H})$.

The gradient dissipation condition (Eq. 5) is that the evolution of the affinities obeys $\dot{\mathbf{X}} = -k\partial\bar{\psi}/\partial\mathbf{X}$ for fixed controls, where $\dot{\mathbf{X}}$ is a second order tensor. By the chain rule,

$$\dot{\mathbf{X}} = \frac{\partial\mathbf{X}}{\partial\mathbf{E}} \dot{\mathbf{E}}. \quad (43)$$

Here $\partial\mathbf{X}/\partial\mathbf{E}$ is a fourth order tensor. Again by the chain rule,

$$\frac{\partial\bar{\psi}}{\partial\mathbf{X}} = \frac{\partial\psi}{\partial\mathbf{E}} \frac{\partial\mathbf{E}}{\partial\mathbf{X}}. \quad (44)$$

Substitution of Eqs. 43 and 44 into $\dot{\mathbf{X}} = -k\partial\bar{\psi}/\partial\mathbf{X}$ yields the evolution equation in terms of the state variables,

$$\dot{\mathbf{E}} = -k \left(\frac{\partial\mathbf{X}}{\partial\mathbf{E}} \right)^{-2} \frac{\partial\psi}{\partial\mathbf{E}} = -k \left(\frac{\partial\mathbf{X}}{\partial\mathbf{E}} \right)^{-2} \left(\frac{\partial\phi}{\partial\mathbf{E}} + \mathbf{H} \right). \quad (45)$$

This form reduces to that of Eq. 6 in the case that $\mathbf{E}$ has only diagonal non-zero entries. Then $\mathbf{H}$ also has only diagonal non-zero entries. The affinity is a second order tensor defined by

$$(\mathbf{X})_{ij} = \left(\frac{\partial\psi}{\partial\mathbf{E}} \right)_{ij} = \frac{\partial\psi}{\partial\mathbf{E}_{ij}}. \quad (46)$$

If ψ only depends on E_{11} , E_{22} , and E_{33} , then the only non-zero terms of $\mathbf{X}$ are the three terms $\mathbf{X}_{ii}$.

The fourth order tensor $\partial\mathbf{X}/\partial\mathbf{E}$ is defined by

$$\left(\frac{\partial\mathbf{X}}{\partial\mathbf{E}} \right)_{ijkm} = \frac{\partial\mathbf{X}_{ij}}{\partial\mathbf{E}_{km}}. \quad (47)$$

The only non-zero terms of this tensor, if only the diagonal entries of $\mathbf{E}$ are non-zero, have the form

$$\left(\frac{\partial\mathbf{X}}{\partial\mathbf{E}} \right)_{iikk} = \frac{\partial\mathbf{X}_{ii}}{\partial\mathbf{E}_{kk}} = \frac{\partial^2\psi}{\partial\mathbf{E}_{ii}\partial\mathbf{E}_{kk}}. \quad (48)$$

The evolution equation therefore has the form of Eq. 6.

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